

Manura Jithmal De Silva

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 LinkedIn: [Manura Jithmal De Silva](#)

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Personal Portfolio: <https://manuradesilva.myportfolio.com/>

Objective

Passionate and motivated graduate researcher with interdisciplinary experience in Mechanical Engineering, Biomechanics, and Rehabilitation Robotics. Seeking opportunities to contribute to research and teaching while pursuing innovations at the intersection of Engineering, Healthcare, and Sports Science.

Education

MSc (Research) in Mechanical Engineering (In Progress)

University of Moratuwa, Sri Lanka

April 2022 – Present

- Thesis: *Development of a Robotic Ankle Exoskeleton for Therapeutic Exercises*
- Focus: Rehabilitation Robotics, Stroke Recovery
- Expected Completion: August, 2025

BSc (Hons) in Mechanical Engineering

University of Moratuwa, Sri Lanka

Graduated: 2019

- Final Year Project: Development of a Wearable Exoskeleton Chair for Industrial Workers
 - Publication in ICORR 2019: [ChairX: a robotic exoskeleton chair for industrial workers](#)
- Team Leader for, “Development of an Endoscopic Scissor Mechanism using a MEMS based Comb - Drive Actuator” - Semester 7 Project under MEMS

- Team Leader for “Development of User following Robot Luggage Carrier” for Semester 3 project of Fundamentals of Mechatronics
 - Reverse Engineering approach of an Angle Grinder with Manufacturing processes for mini project in Semester 3
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Research Experience

Graduate Research Scholar

University of Moratuwa – NIHR UK-Funded Project

Nov 2021 – Apr 2022

- Project: *Development of a Wire Clamp for Circular External Fixators*
- Collaborated with multidisciplinary teams on orthopedic medical device prototyping and testing
- Successfully completed and documented the research project within the funded period

Collaborative Research

Imperial College London (Remote / Joint Research)

Ongoing

- Supporting research efforts on the project “Development of a Lower-Limb Prosthetic Device for Through-Knee Amputees” under joint supervision from Imperial College London, UK and University of Moratuwa, Sri Lanka
-

Teaching Experience

Visiting Lecturer

Informatics Institute of Technology (University of Westminster Affiliate)

2023 – Present

- Modules: *Robotics Principles, Applied Robotics*
- Designed and delivered course content, practicals, and assessments for undergraduate students in Computer Science and Software Engineering
- Worked with
 - MATLAB, Python, Robotics Toolboxes in MATLAB & Python

Visiting Lecturer

Institute of Engineering Technology (IET), Sri Lanka
2022 – 2023

- Module: *Robotics and Industrial Automation*
- Designed and delivered course content, and prepared Semester End Examination Papers for undergraduate students in Production Engineering Technology
- Worked with
 - PLC Systems (“Logo! Soft Comfort” - PLC Simulation Software)
 - MATLAB

Teaching Assistant / Instructor

University of Moratuwa
2022 – Present

- Assisted in modules such as *Biomechanics, Robotics, and Medical Technology*
 - Worked with
 - **Motion Capture Systems (Vicon and Simi Motion),**
 - **EEG Systems (g.tec and OpenBCI)**
 - **EMG Systems (Delsys Bagnoli and Trigno)**
 - **Basic Physiology Teaching System (BIOPAC Student Lab)**
 - **MATLAB Robotics Toolbox**
 - **Python Robotics Toolbox**
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Industrial Experience

Mechanical Engineer (MEP Sector - AEC Industry)

Building Services Engineering Consultants (Pvt.) Ltd
2019 – 2021

- Involved in **MEP Design and Project (Site) Coordination, Supervision & Consultation**
- Gained hands-on experience in **HVAC, Plumbing, Electrical Installations & Layouts, and Project Management**

Automation Engineering Intern

MAS Intimates (Pvt.) Ltd
June 2017 – December 2017

Intern (Pre-University Period)

Alta Vision Solar (Pvt.) Ltd

February 2014 – July 2014

Biomechanics & Sports Science Involvement

Volunteer Biomechanics Analyst

Institute of Sports Medicine, Sri Lanka

2023 – Present

- Worked in **Human Performance Laboratory** using advanced biomechanical tools:
 - **Vicon Motion Capture System with Blue-Trident IMUs**
 - **Delsys Trigno EMG**
 - **AMTI Force Plates**
 - Gait and Sports Performance Analysis
 - (e.g., Cricket Fast Bowling, Weight Lifting etc.)
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Supervision & Mentorship

Final Year Project Co-Supervisor

Department of Mechanical Engineering, University of Moratuwa

2022 – 2024

- Supervised projects in Rehabilitation Robotics, Prosthetics, and Assistive Devices
 - Guiding a student team towards commercialization of a prototype via startup
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Entrepreneurial Activities

Co-Founder & Director

ExoGenic (Pvt.) Ltd

2024 - Present

- Startup focused on commercializing assistive and rehabilitation technologies
 - Currently developing and testing an upper-limb rehab device
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Skills

Engineering & Research:

SolidWorks, MATLAB, Arduino, ROS, ANSYS (basic), LabVIEW,
COMSOL Multiphysics, PLC Programming (Siemens and Yokogawa),

Biomechanics Tools:

Vicon Nexus, Simi Motion, Delsys EMGworks, AMTI Force Plate System,
Blue Trident IMUs

Programming:

Python, C/C++, Arduino IDE

Teaching & Presentation:

Course Design, Lab Demonstrations, Student Mentoring

Soft Skills:

Project Coordination, Interdisciplinary Collaboration, Communication

Certifications & Workshops

Vicon Training (January 2023) - On Vicon Motion Capture Systems

CPD on Heating, Ventilation and Air-Conditioning by IESL (2021)

CPD on Air-Conditioning Systems Design by IESL (2021)

Academic Publications & Conferences

Journal Publications:

- [1] K. N. D. Widanage, M. J. De Silva, T. D. Lalitharatne, A. M. J. Bull, and R. Gopura, "Developments in circular external fixators: A review," *Injury*, vol. 54, no. 12, p. 111157, 2023.

Conference Publications :

- [1] D. Wijegunawardana et al., "ChairX: a robotic exoskeleton chair for industrial workers," in 2019 IEEE 16th International Conference on Rehabilitation Robotics (ICORR), 2019, pp. 587–592.

- [2] W. Weerasoory, U. R. E. Kumasar, H. M. W. Nipun, H. De Silva, R. Ranaweera, and R. Gopura, "A Robotic Hand for Rehabilitation of Wrist and Fingers," in *2023 8th International Conference on Information Technology Research (ICITR)*, 2023, pp. 1–6.
- [3] G. Bandara, D. D. Amarasinghe, H. Hettiarachchi, H. De Silva, R. Ranaweera, and R. Gopura, "Dexterous Robotic Finger with Joint Mobility Modulation for Adaptive Grasping," in *2024 Moratuwa Engineering Research Conference (MERCon)*, 2024, pp. 530–535.
- [4] D. De Silva *et al.*, "Development of a Wheeled Mobility Device to Facilitate Chair Transfers," in *2024 Moratuwa Engineering Research Conference (MERCon)*, 2024, pp. 730–735.
- [5] V. M. Weeramanthrie, B. M. Kulathilake, M. K. D. S. Jayasuriya, H. De Silva, R. Ranaweera, and R. Gopura, "Vision-Based Stair Recognition and Feature Extraction Method for Lower-Limb Wearable Robots," in *2024 Moratuwa Engineering Research Conference (MERCon)*, 2024, pp. 1–6.

Abstracts :

- [1] H.M.J. De Silva, K.N.D. Widanage, O.L.N.P. Dharmaratne, Anthony MJ Bull, RARC Gopura, "Development of a Circular external fixator to enhance the wire tension stability", in *2023 The 12th International Conference on Biomedical Engineering and Biotechnology (ICBEB 2023)*

Magazine Articles :

- [1] K. N. Dickwella Widanage *et al.*, "Will the new circular external fixator ease the surgeon's burden?". *Bolgoda Plains*

Intellectual Property

Patent Applications (Pending):

1. **Wire Clamping Mechanism for Circular External Fixators**
Application Ref. No.: 23311 – National Intellectual Property Office of Sri Lanka
 Filed: 2024
 Applicant(s): R.A.R.C. Gopura, D.W.K. Nima, H.H.M.J. De Silva, O.N.L.P. Dharmaratne

2. **Pin and Socket-Based Joint-Mobility Manipulation of an Underactuated Manipulator**

Application Ref. No.: 23440 – National Intellectual Property Office of Sri Lanka

Filed: 2025

Applicant(s): R.A.R.C. Gopura, R.K.P.S. Ranaweera, H.H.M.J. De Silva,
G.M.H.D.Bandara, D.D. Amarasinghe, H.A.N.P. Hettiarachchi

Extracurricular Activities & Achievements

- Assistant Secretary, IEEE EMBS Sri Lanka Chapter (2024)
 - Treasurer, IEEE EMBS Sri Lanka Chapter (2025)
 - Represented Rahula College Football Team (2007-2009)
 - Secretary of the Dancing Society of University of Moratuwa (2018/2019)
 - Committee member of the Dancing Society of University of Moratuwa (2016/2017)
 - Member of the Organizing Committee for IEEE SS12 IoT Competition Finals - 2016
 - Special Achievements:
 - “Achiever’s Award 2014” from NSB for the outstanding performance at GCE Advanced Level Examination 2013
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Personal Details

Name :	Manura Jithmal De Silva
Date of Birth :	04th October 1993
Nationality :	Sri Lankan
NIC Number :	199327804971
Gender :	Male

Non-related Referees

Professor Ruwan Gopura

Dean, Faculty of Graduate Studies,
Senior Professor, Dept. of Mechanical Eng.,
Faculty of Engineering,
University of Moratuwa,
E-mail: gopura@gmail.com

Dr. Pubudu Ranaweera

Senior Lecturer,
Department of Mechanical Engineering,
Faculty of Engineering,
University of Moratuwa,
E-mail: pubudur@uom.lk

I do here by certify that the information provided above are true and correct to the best of my knowledge.

21 / 07 / 2025

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Date



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Signature